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2 **Supporting Information for**

3 **Semi-parametric robust mean estimation based on provable quantile inequality**

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Supporting Information Text

1. Quantile inequality and symmetric weighted average inequality

Unlike the mean-median-mode inequality, whose necessary and sufficient condition is often hard to compute. The following result shows another advantage of trimming inequality.

Theorem 1.1. *A necessary and sufficient condition of the trimming inequality is the monotonic behavior of the bias of trimmed mean as a function of the breakdown point ϵ .*

Proof. Just considering the right-skewed distributions without loss of generality, from the definition of trimming inequality, since $\forall \epsilon_1 \leq \epsilon_2 \leq \frac{1}{2}$, $TM_{\epsilon_1} \geq TM_{\epsilon_2}$, therefore

$$\frac{TM_{\epsilon_1} - \mu}{\sigma} \geq \frac{TM_{\epsilon_2} - \mu}{\sigma} \iff B_{TM_{\epsilon_1}} \geq B_{TM_{\epsilon_2}} \iff B_{TM}(\epsilon_1) \geq B_{TM}(\epsilon_2),$$

which implies the monotonicity of $B_{TM}(\epsilon)$ with respect to ϵ . $\square$

The bias function is free of scale parameter, so the derivatives are much easier to compute. A useful sufficient condition is the monotonicity of symmetric quantile average with respect to ϵ .

Theorem 1.2. *A sufficient condition of the trimming inequality for a right-skewed distribution is the monotonicity of the symmetric quantile average with respect to the breakdown point ϵ .*

Proof. The trimming inequality is equivalent to, $\forall 0 < \epsilon < \frac{1}{2}$, $\frac{1}{1-2\epsilon+2\delta} \int_{\epsilon-\delta}^{1-\epsilon+\delta} Q(u) du \geq \frac{1}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du$, where δ is an infinitesimal quantity.

$$\begin{aligned} & \text{Then, by deducing } \frac{1}{1-2\epsilon+2\delta} \int_{\epsilon-\delta}^{1-\epsilon+\delta} Q(u) du = \frac{1}{\frac{1}{2}-\epsilon+\delta} \int_{\epsilon-\delta}^{\frac{1}{2}} SQA(u) du = \lim_{n \rightarrow \infty} \left(\frac{1}{\left(\frac{1}{2}n-n\epsilon+1\right)} \sum_{i=n\epsilon-1}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} \right) \\ & > \lim_{n \rightarrow \infty} \left(\frac{1}{\left(\frac{1}{2}n-n\epsilon+1\right)} \left(\left(\sum_{i=n\epsilon}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} \right) + \frac{X_{n\epsilon}+X_{n-n\epsilon}}{2} \right) \right) \\ & = \lim_{n \rightarrow \infty} \left(\frac{1}{\left(\frac{1}{2}n-n\epsilon\right)} \left(\frac{\left(\frac{1}{2}n-n\epsilon\right)}{\left(\frac{1}{2}n-n\epsilon+1\right)} \left(\sum_{i=n\epsilon+1}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} \right) + \frac{2\left(\frac{1}{2}n-n\epsilon\right)}{\left(\frac{1}{2}n-n\epsilon+1\right)} \frac{X_{n\epsilon}+X_{n-n\epsilon}}{2} \right) \right) \\ & = \lim_{n \rightarrow \infty} \left(\frac{1}{\left(\frac{1}{2}n-n\epsilon\right)} \left(\frac{\left(\frac{1}{2}n-n\epsilon\right)}{\left(\frac{1}{2}n-n\epsilon+1\right)} \left(\sum_{i=n\epsilon+1}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} \right) + \frac{X_{n\epsilon}+X_{n-n\epsilon}}{2} + \frac{\left(\frac{1}{2}n-n\epsilon-1\right)}{\left(\frac{1}{2}n-n\epsilon+1\right)} \frac{X_{n\epsilon}+X_{n-n\epsilon}}{2} \right) \right) \\ & > \lim_{n \rightarrow \infty} \left(\frac{1}{\frac{1}{2}n-n\epsilon} \left(\frac{\frac{1}{2}n-n\epsilon}{\frac{1}{2}n-n\epsilon+1} \sum_{i=n\epsilon+1}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} + \frac{X_{n\epsilon}+X_{n-n\epsilon}}{2} + \frac{1}{\frac{1}{2}n-n\epsilon+1} \left(\frac{X_{n\epsilon+1}+X_{n-n\epsilon-1}}{2} + \dots + \frac{X_{\frac{1}{2}n}+X_{\frac{1}{2}n}}{2} \right) \right) \right) \\ & = \lim_{n \rightarrow \infty} \left(\frac{1}{\left(\frac{1}{2}n-n\epsilon\right)} \left(\left(\sum_{i=n\epsilon}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} \right) \right) \right) = \frac{1}{1-2\epsilon} \int_{\epsilon}^{1-\epsilon} Q(u) du, \text{ the proof is complete. } \square \end{aligned}$$

Theorem 1.3. *A necessary and sufficient condition of the quantile inequality is the monotonic behavior of the bias of the symmetric quantile average as a function of the breakdown point ϵ .*

Proof. The proof is analogous to Theorem 1.1, just replacing the TM as SQA. $\square$

Quantile inequality is more fundamental than trimming inequality; for example, quantile inequality also implies Winsorization inequality.

Definition 1.1 (Winsorization inequality). A distribution follows the Winsorization inequality if and only if $\forall \epsilon_1 \leq \epsilon_2 \leq \frac{1}{2}$, $WM_{\epsilon_1} \geq WM_{\epsilon_2}$, or $\forall \epsilon_1 \leq \epsilon_2 \leq \frac{1}{2}$, $WM_{\epsilon_1} \leq WM_{\epsilon_2}$.

Theorem 1.4. *A sufficient condition of the Winsorization inequality for a right-skewed distribution is the monotonicity of the symmetric quantile average function with respect to the breakdown point ϵ .*

Proof. The Winsorization inequality in Definition 1.1 is equivalent to,

$\forall 0 < \epsilon < \frac{1}{2}$, $\int_{\epsilon-\delta}^{1-\epsilon+\delta} Q(u) du + (\epsilon - \delta)(Q(\epsilon - \delta) + Q(1 - \epsilon + \delta)) \geq \int_{\epsilon}^{1-\epsilon} Q(u) du + \epsilon(Q(\epsilon) + Q(1 - \epsilon))$, where δ is an infinitesimal quantity.

$$\begin{aligned} & \text{Then, by deducing } \int_{\epsilon-\delta}^{1-\epsilon+\delta} Q(u) du + (\epsilon - \delta)(Q(\epsilon - \delta) + Q(1 - \epsilon + \delta)) = \int_{\epsilon-\delta}^{\frac{1}{2}} SQA(u) du + 2(\epsilon - \delta) SQA(\epsilon - \delta) \\ & = \lim_{n \rightarrow \infty} \left(\frac{2}{n} \left(\sum_{i=n\epsilon-1}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} + (n\epsilon-1) \left(\frac{X_{n\epsilon-1}+X_{n-n\epsilon+1}}{2} \right) \right) \right) \\ & > \lim_{n \rightarrow \infty} \left(\frac{2}{n} \left(\sum_{i=n\epsilon}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} + \frac{X_{n\epsilon}+X_{n-n\epsilon}}{2} + (n\epsilon-1) \left(\frac{X_{n\epsilon-1}+X_{n-(n\epsilon-1)}}{2} \right) \right) \right) \\ & = \lim_{n \rightarrow \infty} \left(\frac{2}{n} \left(\sum_{i=n\epsilon}^{\frac{1}{2}n} \frac{X_i+X_{n-i}}{2} + (n\epsilon) \left(\frac{X_{n\epsilon}+X_{n-n\epsilon}}{2} \right) \right) \right) \\ & = \int_{\epsilon}^{1-\epsilon} Q(u) du + (\epsilon)(Q(\epsilon) + Q(1 - \epsilon)), \text{ the proof is complete. } \square \end{aligned}$$

Theorem 1.5. *A necessary and sufficient condition of the second quantile inequality is the monotonicity and convexity of the bias function of the symmetric quantile average with respect to the breakdown point ϵ .*

Proof. The proof is analogous to Theorem 1.1. $\square$

For simplicity, $0 < \epsilon < \frac{1}{2}$ is assumed in the following proofs unless otherwise specified.

Theorem 1.6. The Weibull distribution follows the quantile inequality if the shape parameter $\alpha \leq \frac{1}{1-\ln(2)} \approx 3.259$.

Proof. The pdf of the Weibull distribution is $f(x) = \frac{\alpha e^{-(\frac{x}{\lambda})^\alpha}}{\lambda} (\frac{x}{\lambda})^{\alpha-1}$, $x \geq 0$, the quantile function is $F^{-1}(p) = \lambda(-\ln(1-p))^{1/\alpha}$, $1 \geq p \geq 0$, $\alpha > 0$, $\lambda > 0$. Then, the standardized bias of a symmetric quantile average with a breakdown point ϵ , is

$$B_{\text{SQA}}(\epsilon, \alpha) = \frac{\text{SQA}_\epsilon - \mu}{\sigma} = \frac{\frac{1}{2}(\lambda(-\ln(1-\epsilon))^{1/\alpha} + \lambda(-\ln(\epsilon))^{1/\alpha}) - \lambda\Gamma(1 + \frac{1}{\alpha})}{\sqrt{\lambda^2 \left(\Gamma(1 + \frac{2}{\alpha}) - \Gamma(1 + \frac{1}{\alpha})^2 \right)}}.$$

$\frac{\partial B_{\text{SQA}}}{\partial \epsilon} = \frac{\frac{(-\ln(1-\epsilon))^{\frac{1}{\alpha}-1}}{\alpha-\alpha\epsilon} + \frac{(-\ln(\epsilon))^{1/\alpha}}{\alpha\epsilon \ln(\epsilon)}}{2\sqrt{\Gamma(\frac{\alpha+2}{\alpha}) - \Gamma(1+\frac{1}{\alpha})^2}}$. Let $g(\epsilon, \alpha) = \frac{(-\ln(1-\epsilon))^{\frac{1}{\alpha}-1}}{\alpha-\alpha\epsilon} + \frac{(-\ln(\epsilon))^{1/\alpha}}{\alpha\epsilon \ln(\epsilon)} = -(-\ln(1-\epsilon))^{\frac{1}{\alpha}}((1-\epsilon)(\ln(1-\epsilon)))^{-1} + (-\ln(\epsilon))^{\frac{1}{\alpha}}(\epsilon \ln(\epsilon))^{-1}$. Arranging the equation $g(\epsilon, \alpha) = 0$, it can be shown that $\frac{\epsilon}{(1-\epsilon)} = \left(\frac{\ln(\epsilon)}{\ln(1-\epsilon)}\right)^{\frac{1}{\alpha}-1}$. Let $L(\epsilon) = \frac{\epsilon}{(1-\epsilon)}$, $R(\epsilon, \alpha) = \left(\frac{\ln(\epsilon)}{\ln(1-\epsilon)}\right)^{\frac{1}{\alpha}-1}$, $LmR(\epsilon, \alpha) = L(\epsilon, \alpha) - R(\epsilon, \alpha)$, then $\frac{\partial LmR}{\partial \alpha} = \frac{\ln\left(\frac{\ln(\epsilon)}{\ln(1-\epsilon)}\right)\left(\frac{\ln(\epsilon)}{\ln(1-\epsilon)}\right)^{\frac{1}{\alpha}-1}}{\alpha^2}$. For $0 < \epsilon < \frac{1}{2}$, $\frac{\partial LmR}{\partial \alpha} > 0$, so $LmR(\epsilon, \alpha)$ is monotonic with respect to α . When $\alpha = \frac{1}{1-\ln(2)}$, $g(\epsilon) = -\frac{1}{\epsilon(-\ln(\epsilon))^{\ln(2)}} + \frac{1}{(1-\epsilon)(-\ln(1-\epsilon))^{\ln(2)}}$. Let $h(\epsilon) = \epsilon(-\ln(\epsilon))^{\ln(2)}$, $h'(\epsilon) = \frac{(-\ln(\epsilon))^{\ln(2)} \ln(2\epsilon)}{\ln(\epsilon)}$, for $0 < \epsilon < e^{-\ln(2)} = \frac{1}{2}$, $h'(\epsilon) > 0$. As a result, $h(\epsilon)$ is monotonic increasing, $-h(1-\epsilon)$ is monotonic increasing, $h(\epsilon) - h(1-\epsilon)$ is also monotonic increasing. So, if $0 < \epsilon < \frac{1}{2}$, $h(\epsilon) - h(1-\epsilon) < h(\frac{1}{2}) - h(1-\frac{1}{2}) = 0$, $g(\epsilon, \alpha) < 0$. So, $\frac{\partial B_{\text{SQA}}}{\partial \epsilon} < 0$, $B_{\text{SQA}}(\epsilon, \alpha)$ is monotonic decreasing in ϵ when $\alpha \leq \frac{1}{1-\ln(2)}$. The assertion follows from Theorem 1.3. $\square$

Since the exponential and Rayleigh distributions are special cases of the Weibull distribution when $\alpha = 1$ and $\alpha = 2$, they also follow the quantile inequality.

Remark. The Weibull distribution can be symmetric. Its skewness is $\tilde{\mu}_3 = \frac{2\Gamma(1+\frac{1}{\alpha})^3 - 3\Gamma(1+\frac{2}{\alpha})\Gamma(1+\frac{1}{\alpha}) + \Gamma(1+\frac{3}{\alpha})}{(\Gamma(1+\frac{2}{\alpha}) - \Gamma(1+\frac{1}{\alpha})^2)^{3/2}}$. Denote the solution of $\tilde{\mu}_3 = 0$ as $\alpha_0 \approx 3.602$. The above proof implies that when α is close to α_0 , the bias function of SQA is no longer monotonic. The bias function of the trimmed mean for the Weibull distribution is

$$B_{\text{TM}}(\epsilon, \alpha) = \frac{\frac{\Gamma(1+\frac{1}{\alpha}, -\ln(1-\epsilon)) - \Gamma(1+\frac{1}{\alpha}, -\ln(\epsilon))}{1-2\epsilon} - \Gamma(1+\frac{1}{\alpha})}{\sqrt{\Gamma(1+\frac{2}{\alpha}) - \Gamma(1+\frac{1}{\alpha})^2}},$$

Despite the fact that Theorem 1.2 is not necessary, numerical results show, when $\alpha = 3.259$, $\epsilon = 0.1$, $\frac{\partial B_{\text{TM}}}{\partial \epsilon} \approx -0.164$, when $\epsilon = 0.4$, $\frac{\partial B_{\text{TM}}}{\partial \epsilon} \approx -0.001$, when $\epsilon = 0.499$, $\frac{\partial B_{\text{TM}}}{\partial \epsilon} \approx 7.66 \times 10^{-8}$. So, the bias function of the trimmed mean can also be non-monotonic when $\alpha \geq \frac{1}{1-\ln(2)}$. A plot of $\frac{\partial B_{\text{TM}}}{\partial \epsilon}$ for $0 < \epsilon < 0.5$ is given in Figure S1.

Additionally, the second derivative of the bias function of SQA for the Weibull distribution is,

$$\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon} = \frac{-\frac{(-\ln(\epsilon))^{\frac{1}{\alpha}-1}}{\alpha^2 \epsilon^2 \ln(\epsilon)} - \frac{(-\ln(\epsilon))^{1/\alpha}}{\alpha \epsilon^2 \ln^2(\epsilon)} - \frac{(-\ln(\epsilon))^{1/\alpha}}{\alpha \epsilon^2 \ln(\epsilon)} + \frac{(\frac{1}{\alpha}-1)(-\ln(1-\epsilon))^{\frac{1}{\alpha}-2}}{(1-\epsilon)(\alpha-\alpha\epsilon)} + \frac{\alpha(-\ln(1-\epsilon))^{\frac{1}{\alpha}-1}}{(\alpha-\alpha\epsilon)^2}}{2\sqrt{\Gamma(\frac{\alpha+2}{\alpha}) - \Gamma(1+\frac{1}{\alpha})^2}}. \text{ The numerical solutions show that}$$

when $\alpha < 3$, $\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon} > 0$, if $0 < \epsilon < \frac{1}{2}$. The flip of the signs also occurs when α is close to $\frac{1}{1-\ln(2)} \approx 3.259$. When $\alpha = \frac{1}{1-\ln(2)}$, $\epsilon = 0.1$, $\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon} \approx 3.259$, when $\epsilon = 0.4$, $\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon} \approx 0.020$, when $\epsilon = 0.5$, $\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon} \approx -7.34 \times 10^{-16}$. A plot of $\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon}$ for $0.25 < \epsilon < 0.5$ is given in Figure S1.

Then, the third derivative of the SQA for the Weibull distribution is,

$$\frac{\partial^3 B_{\text{SQA}}}{\partial^3 \epsilon} = -\frac{(\alpha-1)(2\alpha-1)(\epsilon-1)^3(-\ln(\epsilon))^{\frac{1}{\alpha}-3} - 3(\alpha-1)\alpha(\epsilon-1)^3(-\ln(\epsilon))^{\frac{1}{\alpha}-2}}{2\alpha^3(\epsilon-1)^3\epsilon^3\sqrt{\Gamma(\frac{\alpha+2}{\alpha}) - \Gamma(1+\frac{1}{\alpha})^2}} - \frac{2\alpha^2\epsilon^3(-\ln(1-\epsilon))^{\frac{1}{\alpha}-1} + 2\alpha^2(\epsilon-1)^3(-\ln(\epsilon))^{\frac{1}{\alpha}-1} + (1-\alpha)(1-2\alpha)\epsilon^3(-\ln(1-\epsilon))^{\frac{1}{\alpha}-3} + 3(1-\alpha)\alpha\epsilon^3(-\ln(1-\epsilon))^{\frac{1}{\alpha}-2}}{2\alpha^3(\epsilon-1)^3\epsilon^3\sqrt{\Gamma(\frac{\alpha+2}{\alpha}) - \Gamma(1+\frac{1}{\alpha})^2}}. \text{ The numerical solutions show}$$

that when $\alpha < 3$, $\frac{\partial^3 B_{\text{SQA}}}{\partial^3 \epsilon} < 0$, if $0 < \epsilon < \frac{1}{2}$. The flip of the signs occurs when $\alpha = 3.471$. When $\epsilon = 0.4$, $\frac{\partial^3 B_{\text{SQA}}}{\partial^3 \epsilon} \approx -0.0218$, when $\epsilon = 0.499$, $\frac{\partial^3 B_{\text{SQA}}}{\partial^3 \epsilon} \approx 4.928 \times 10^{-5}$. A plot of $\frac{\partial^3 B_{\text{SQA}}}{\partial^3 \epsilon}$ for $0.3 < \epsilon < 0.5$ is given in Figure S1.

In addition, because the kurtosis range used here for the Weibull distribution starts from 3.1, the corresponding α is 2.133, so among the kurtosis range discussed here, the numerical results show that the Weibull distribution follows the first three quantile inequalities (Figure S2).

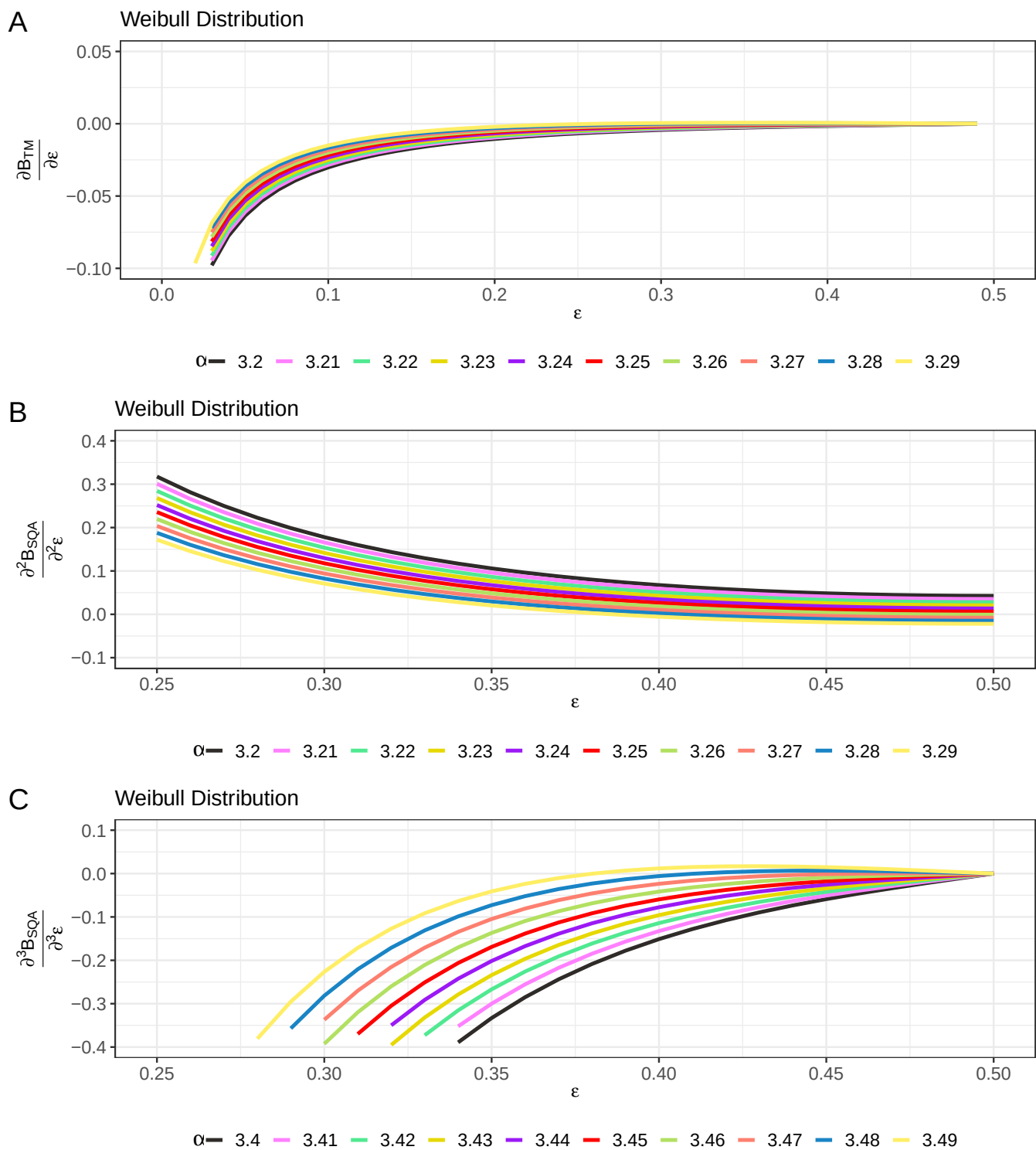
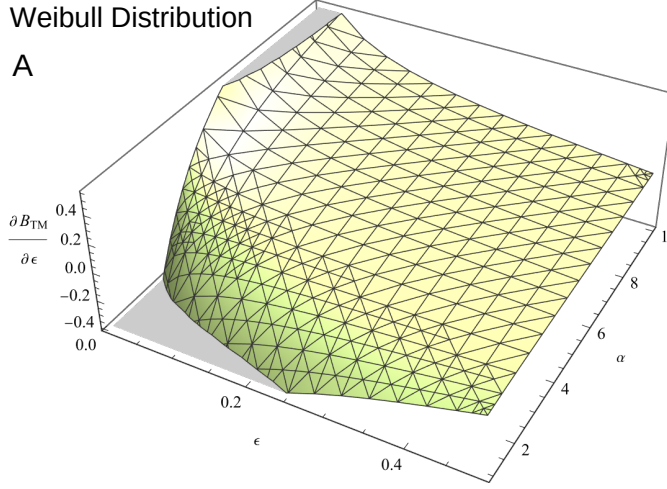


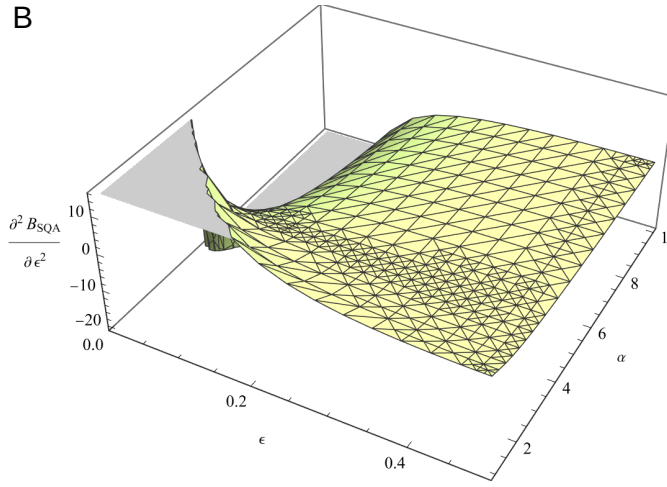
Fig. S1. A. The first derivative of the bias function of TM for the Weibull distribution as a function of the breakdown point ϵ . B. The second derivative of the bias function of SQA for the Weibull distribution as a function of the breakdown point ϵ . C. The third derivative of the bias function of SQA for the Weibull distribution as a function of the breakdown point ϵ .

Weibull Distribution

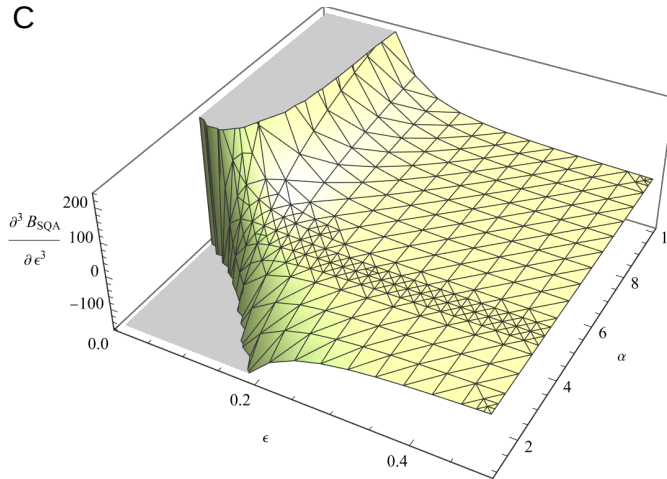
A



B

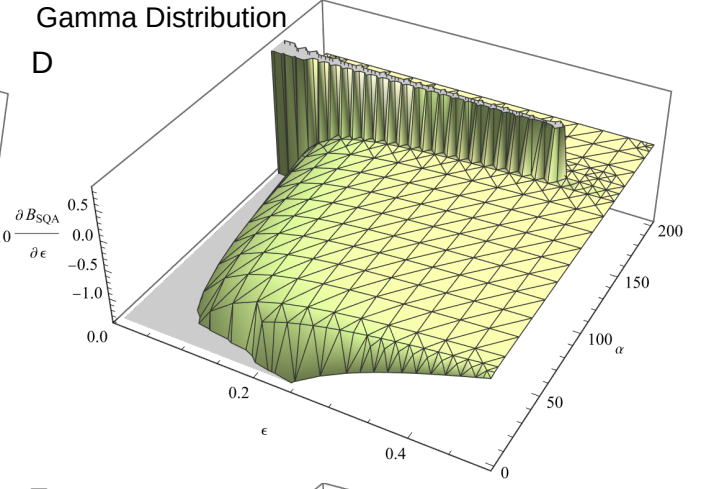


C

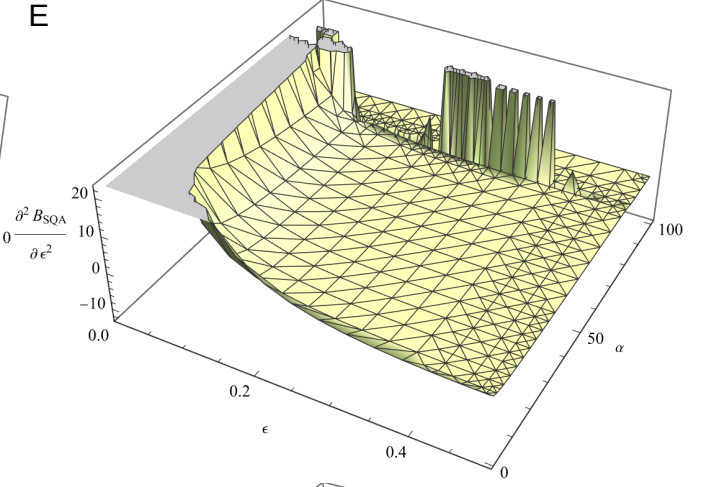


Gamma Distribution

D



E



F

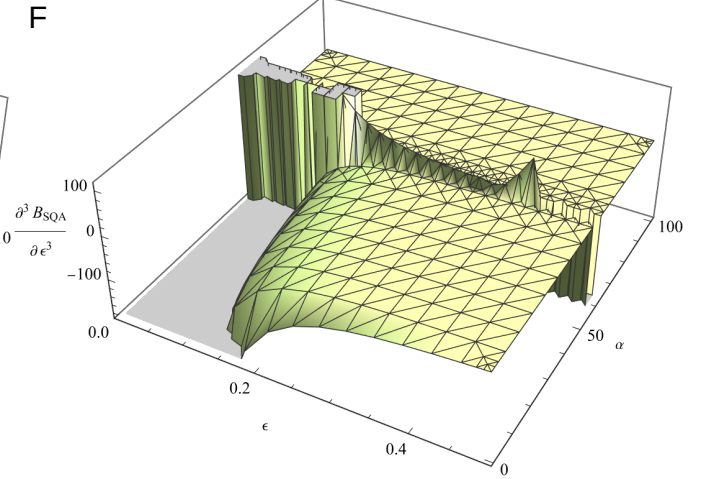


Fig. S2. 3D plots of the derivatives of the bias function of SQA or TM, the left is for the Weibull distribution, the right is for the gamma distribution. The derivatives of the bias functions for the Weibull distribution are continuous over the domain. In contrast, those of the SQA bias function for the gamma distribution are continuous before the flips of signs occur.

The pdf of the gamma distribution is $f(x) = \frac{\lambda^{-\alpha} x^{\alpha-1} e^{-\frac{x}{\lambda}}}{\Gamma(\alpha)}$, $x \geq 0$, the quantile function is $Q(p) = \lambda P^{-1}(\alpha, p)$, $1 \geq p \geq 0$, $\alpha > 0, \lambda > 0$, P is the regularized incomplete gamma function. So, $E[X] = \int_{-\infty}^{\infty} x f(x) dx = \alpha\lambda$. Similarly, the variance is $\alpha\lambda^2$. Then, the standardized bias of a symmetric quantile average with a breakdown point ϵ , is

$$B_{\text{SQA}}(\epsilon, \alpha) = \frac{\text{SQA}_{\epsilon} - \mu}{\sigma} = \frac{\frac{1}{2}(\lambda P^{-1}(\alpha, 1 - \epsilon) + \lambda P^{-1}(\alpha, \epsilon)) - \alpha\lambda}{\sqrt{\alpha\lambda^2}}.$$

$\frac{\partial B_{\text{SQA}}}{\partial \epsilon} = \frac{\Gamma(\alpha)}{2\sqrt{\alpha}}(e^{P^{-1}(\alpha, \epsilon)} P^{-1}(\alpha, \epsilon)^{1-\alpha} - e^{P^{-1}(\alpha, 1-\epsilon)} P^{-1}(\alpha, 1-\epsilon)^{1-\alpha})$. It is trivial to show that when $\alpha \leq 1$, $P^{-1}(\alpha, \epsilon)$ is monotonic increasing in ϵ , if $0 < \epsilon < \frac{1}{2}$. Then $\frac{\partial B_{\text{SQA}}}{\partial \epsilon} < 0$, $B_{\text{SQA}}(\epsilon, \alpha)$ is monotonic decreasing in ϵ over the interval $(0, \frac{1}{2})$. However, the analytical analysis of $\alpha > 1$ is hard since the regularized incomplete gamma function is very complex (1-3). Numerical results shows that the flip of signs of $\frac{\partial B_{\text{SQA}}}{\partial \epsilon}$ occurs when $\alpha \approx 139.5$, Figure S3. The second derivative of the bias function for the gamma distribution is $\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon} = \frac{\Gamma(\alpha)^2}{2\sqrt{\alpha}}((1-\alpha)e^{2P^{-1}(\alpha, 1-\epsilon)} P^{-1}(\alpha, 1-\epsilon)^{1-2\alpha} + e^{2P^{-1}(\alpha, 1-\epsilon)} P^{-1}(\alpha, 1-\epsilon)^{2-2\alpha} + (1-\alpha)e^{2P^{-1}(\alpha, \epsilon)} P^{-1}(\alpha, \epsilon)^{1-2\alpha} + e^{2P^{-1}(\alpha, \epsilon)} P^{-1}(\alpha, \epsilon)^{2-2\alpha})$. The flip of signs of $\frac{\partial^2 B_{\text{SQA}}}{\partial^2 \epsilon}$ occurs when $\alpha \approx 78$. The third derivative is much more cumbersome; numerical results show that the flip of signs of $\frac{\partial^3 B_{\text{SQA}}}{\partial^3 \epsilon}$ occurs when $\alpha \approx 55$, Figure S3. Since the kurtosis range of the gamma distribution starts from 3.1, the corresponding α is 60. The second point is $\alpha = 30$. Besides the first point, the numerical results show that the gamma distribution follows the first three quantile inequalities within the kurtosis setting here (Figure S2).

For the lognormal distribution, the pdf of it is $f(x) = \frac{e^{-\frac{(\ln(x)-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi\sigma x}}$, $x \geq 0$, the quantile function is $Q(p) = e^{\mu - \sqrt{2}\sigma \text{erfc}^{-1}(2p)}$, $1 \geq p \geq 0$, $\alpha > 0, \lambda > 0$. So, $E[X] = \int_{-\infty}^{\infty} x f(x) dx = e^{\mu + \frac{\sigma^2}{2}}$. Similarly, the variance is $(e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$. Then, the standardized bias of a symmetric quantile average with a breakdown point ϵ , is

$$B_{\text{SQA}}(\epsilon, \alpha) = \frac{\text{SQA}_{\epsilon} - \mu}{\sigma} = \frac{\frac{1}{2}(e^{\mu - \sqrt{2}\sigma \text{erfc}^{-1}(2\epsilon)} + e^{\mu - \sqrt{2}\sigma \text{erfc}^{-1}(2(1-\epsilon))}) - e^{\mu + \frac{\sigma^2}{2}}}{\sqrt{(e^{\sigma^2} - 1)e^{2\mu + \sigma^2}}}.$$

Since the first two quantile inequalities for the lognormal distribution were already discussed in the Main Text, only $\frac{\partial^3 B_{\text{SQA}}}{\partial^3 \epsilon}$ is plotted here (Figure S4).

2. Bias bound

As stated at the beginning of the Main Text, the bias bound is the first approach to study the bias of a robust location estimator. Bernard et al. (2020) (4) derived the bias bound of the symmetric quantile average for $\mathcal{P}_U$,

$$B_{\text{SQAB}}(\epsilon) = \begin{cases} \frac{1}{2} \left(\sqrt{\frac{4}{9\epsilon} - 1} + \sqrt{\frac{3\epsilon}{4-3\epsilon}} \right) & \frac{1}{6} \geq \epsilon \geq 0 \\ \frac{1}{2} \left(\sqrt{\frac{1-\epsilon}{\epsilon + \frac{1}{3}}} + \sqrt{\frac{3\epsilon}{4-3\epsilon}} \right) & \frac{1}{2} \geq \epsilon > \frac{1}{6}. \end{cases}$$

Theorem 2.1. The above bias bound function, $B_{\text{SQAB}}(\epsilon)$, is monotonic decreasing over the interval $(0, \frac{1}{2})$.

Proof. When $\frac{1}{6} \geq \epsilon \geq 0$, $B'_{\text{SQAB}}(\epsilon) = \frac{1}{(4-3\epsilon)^2 \sqrt{\frac{\epsilon}{12-9\epsilon}}} - \frac{1}{3\sqrt{\frac{4}{9\epsilon} - 1}} = 0$ has three solutions, $\frac{1}{3} \left(-\sqrt[3]{2(2-\sqrt{2})} - \frac{2^{2/3}}{\sqrt[3]{2-\sqrt{2}}} \right)$, $\frac{(1+i\sqrt{3})\sqrt[3]{2-\sqrt{2}}}{3 \cdot 2^{2/3}} + \frac{1-i\sqrt{3}}{3\sqrt[3]{2(2-\sqrt{2})}}$, $\frac{(1-i\sqrt{3})\sqrt[3]{2-\sqrt{2}}}{3 \cdot 2^{2/3}} + \frac{1+i\sqrt{3}}{3\sqrt[3]{2(2-\sqrt{2})}}$, all are not in $(0, \frac{1}{6})$. When $\frac{1}{2} \geq \epsilon > \frac{1}{6}$, $B'_{\text{SQAB}}(\epsilon) = \frac{1}{(4-3\epsilon)^2 \sqrt{\frac{\epsilon}{12-9\epsilon}}} - \frac{1}{(3\epsilon+1)^2 \sqrt{\frac{1-\epsilon}{9\epsilon+3}}} = 0$ has three solutions, $\frac{1}{2}$, $\frac{1}{18}(9-5\sqrt{3})$, $\frac{1}{18}(5\sqrt{3}+9)$, all are not in $(\frac{1}{6}, \frac{1}{2})$. The first and second formula, when $\epsilon = \frac{1}{6}$, are all equal to $\frac{1}{2} \left(\sqrt{\frac{5}{3}} + \frac{1}{\sqrt{7}} \right)$. Hence, if $0 < \epsilon < \frac{1}{2}$, $B'_{\text{SQAB}}(\epsilon) < 0$, which leads to the assertion of this theorem. $\square$

They also investigated the bias bounds of Range Value at Risk (5), which is

$$\text{RVaR}_{\alpha, \beta} = \frac{1}{\beta - \alpha} \int_{\alpha}^{\beta} \text{VaR}(u) du, 0 < \alpha < \beta < 1,$$

where $\text{VaR}(u) = \inf\{x \in \mathbb{R} : F_U(x) \geq u\}$. They pointed out that $\text{VaR}(u)$ is the quantile function, and it is obvious here that $\text{RVaR}_{\alpha, \beta}$ is the trimmed mean. Let $\alpha = \beta = \epsilon$, then

$$\frac{E(\text{TM}_{\epsilon}) - \mu_P}{\sigma_P} \leq \frac{\sup_{P \in \mathcal{P}_U}(T_{\epsilon}) - \mu_P}{\sigma_P} = \frac{\sup_{P \in \mathcal{P}_U}(\text{RVaR}_{\alpha=\epsilon, \beta=\epsilon}) - \mu_P}{\sigma_P} = B_{\text{TB}}(\epsilon),$$

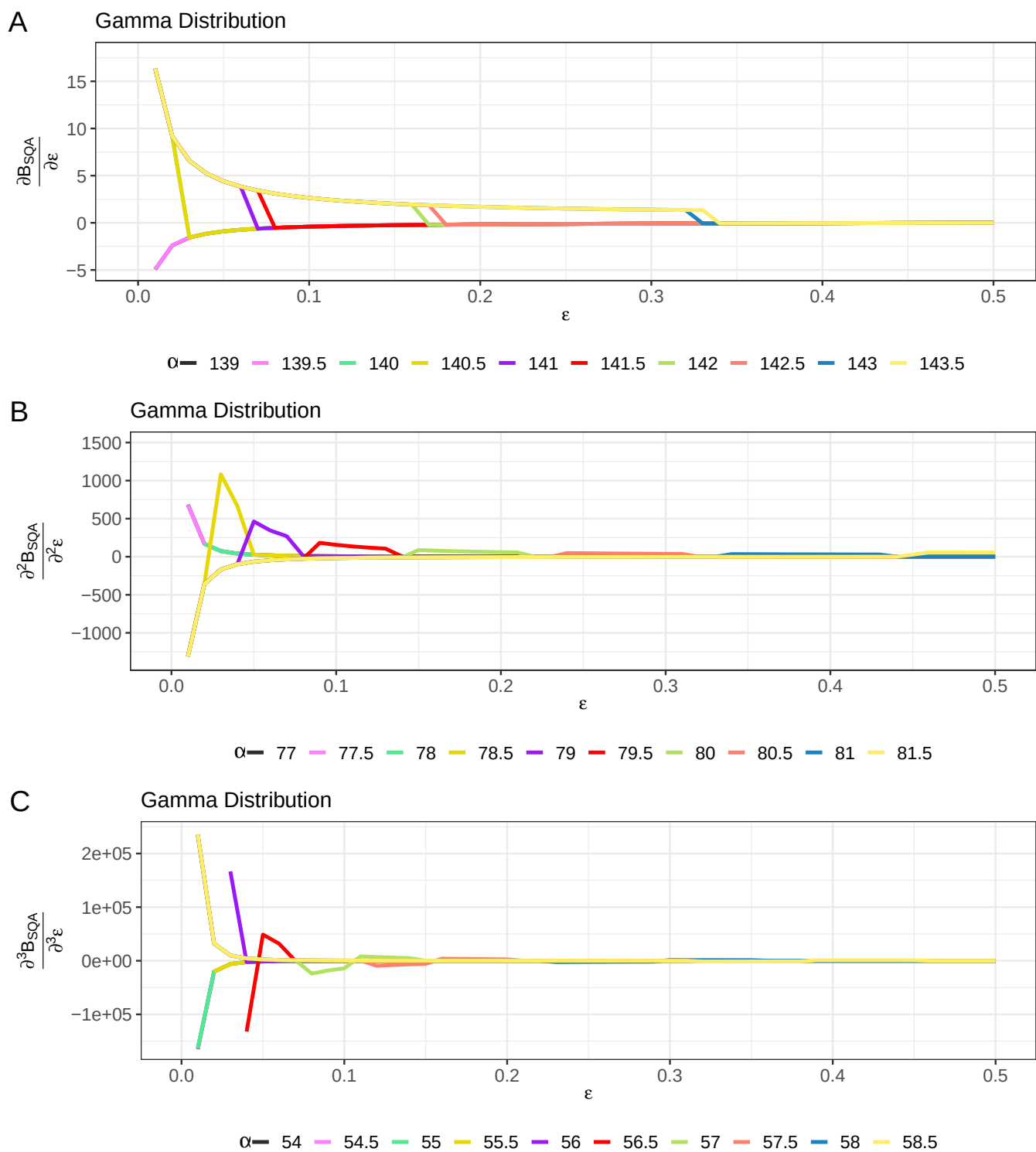


Fig. S3. A. The first derivative of the bias function of SQA for the Gamma distribution as a function of the breakdown point ϵ . B. The second derivative of the bias function of SQA for the gamma distribution as a function of the breakdown point ϵ . C. The third derivative of the bias function of SQA for the gamma distribution as a function of the breakdown point ϵ .

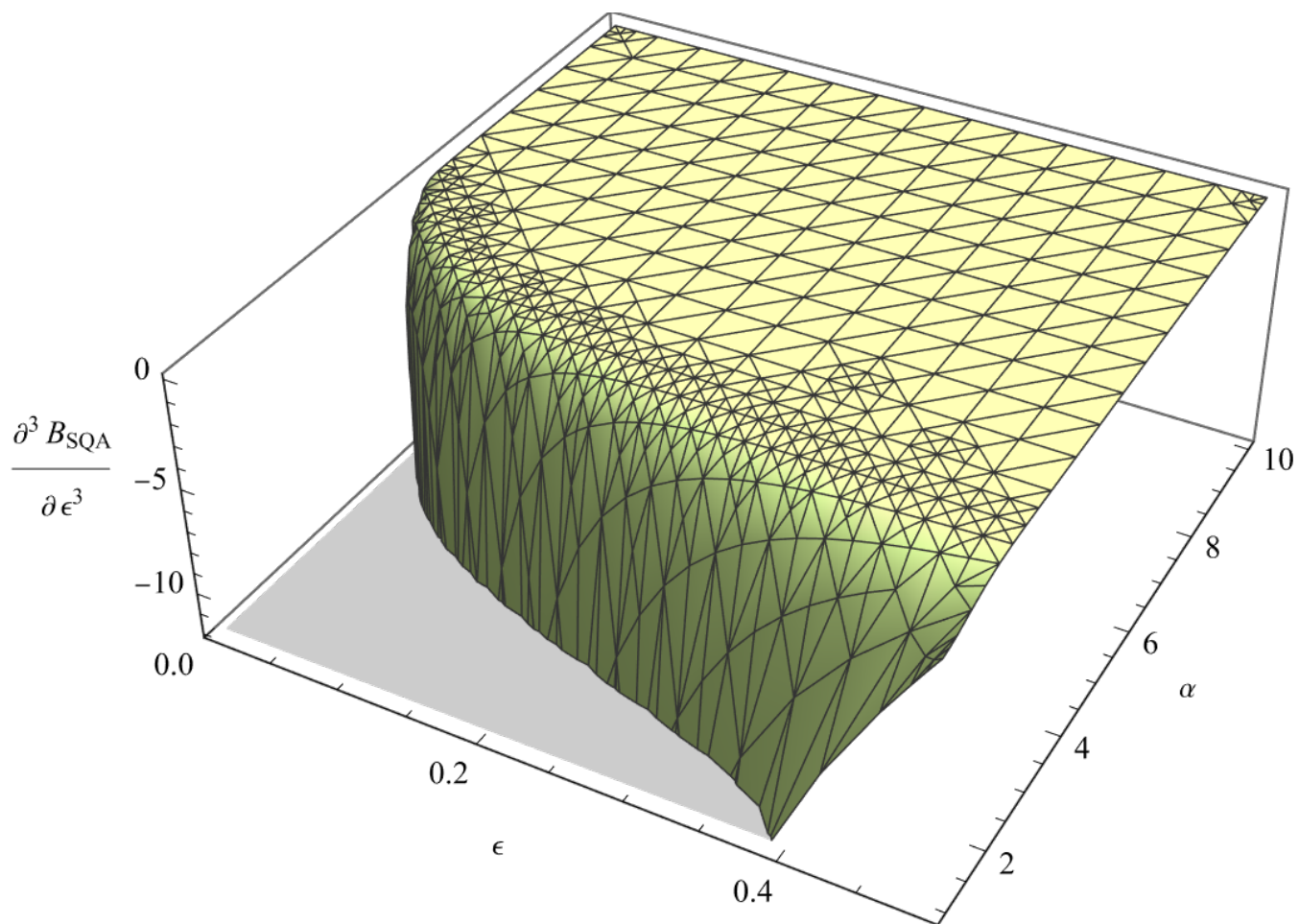


Fig. S4. The third partial derivative of the bias function of SQA for the lognormal distribution.

where

$$B_{TB}(\epsilon) = \frac{\epsilon(9\epsilon^2 + (4 - 3\sqrt{9\epsilon^2 + 4})\epsilon - \sqrt{9\epsilon^2 + 4} + 2)}{(2\epsilon - 1)\sqrt{-\frac{81\epsilon^4}{2} + 3(4\sqrt{9\epsilon^2 + 4} - 9)\epsilon^2 + 6(\sqrt{9\epsilon^2 + 4} - 2)\epsilon + \frac{4}{3}(\sqrt{9\epsilon^2 + 4} - 2) + \frac{9}{2}(3\sqrt{9\epsilon^2 + 4} - 8)\epsilon^3}}.$$

For its full justification, the reader is referred to (4, 6, 7).

An important result of the bias bound of the trimmed mean is that it is monotonic.

Theorem 2.2. *The above bias bound function, $B_{TB}(\epsilon)$, is monotonic increasing over the interval $(0, \frac{1}{2})$.*

$$\text{Proof. Since } B'_{TB}(\epsilon) = \frac{2\sqrt{6}(-5832\epsilon^7 + 232(\sqrt{9\epsilon^2 + 4} - 2)\epsilon + 32(\sqrt{9\epsilon^2 + 4} - 2) + 324(6\sqrt{9\epsilon^2 + 4} - 25)\epsilon^6)}{(1 - 2\epsilon)^2\sqrt{9\epsilon^2 + 4}(-243\epsilon^4 + 18(4\sqrt{9\epsilon^2 + 4} - 9)\epsilon^2 + 36(\sqrt{9\epsilon^2 + 4} - 2)\epsilon + 8(\sqrt{9\epsilon^2 + 4} - 2) + 27(3\sqrt{9\epsilon^2 + 4} - 8)\epsilon^3)^{3/2} +} \\ \frac{2\sqrt{6}(2(397\sqrt{9\epsilon^2 + 4} - 830)\epsilon^2 + 54(50\sqrt{9\epsilon^2 + 4} - 171)\epsilon^5 + 9(294\sqrt{9\epsilon^2 + 4} - 779)\epsilon^4 + 9(193\sqrt{9\epsilon^2 + 4} - 444)\epsilon^3)}{(1 - 2\epsilon)^2\sqrt{9\epsilon^2 + 4}(-243\epsilon^4 + 18(4\sqrt{9\epsilon^2 + 4} - 9)\epsilon^2 + 36(\sqrt{9\epsilon^2 + 4} - 2)\epsilon + 8(\sqrt{9\epsilon^2 + 4} - 2) + 27(3\sqrt{9\epsilon^2 + 4} - 8)\epsilon^3)^{3/2}}.$$

Let $g(\epsilon) = -5832\epsilon^7 + 2(397\sqrt{9\epsilon^2 + 4} - 830)\epsilon^2 + 232(\sqrt{9\epsilon^2 + 4} - 2)\epsilon + 32(\sqrt{9\epsilon^2 + 4} - 2) + 324(6\sqrt{9\epsilon^2 + 4} - 25)\epsilon^6 + 54(50\sqrt{9\epsilon^2 + 4} - 171)\epsilon^5 + 9(294\sqrt{9\epsilon^2 + 4} - 779)\epsilon^4 + 9(193\sqrt{9\epsilon^2 + 4} - 444)\epsilon^3$ and $h(\epsilon)$ denotes the common denominator of $B'_{TB}(\epsilon)$. Then, for $0 < \epsilon < \frac{1}{2}$, $h(\epsilon) > 0$.

$$\text{Let, } g'(\epsilon) = -\frac{2(27(855\sqrt{9\epsilon^2 + 4} - 2158)\epsilon^4 + 9(1558\sqrt{9\epsilon^2 + 4} - 3543)\epsilon^3)}{\sqrt{9\epsilon^2 + 4}} \\ - \frac{2(-61236\epsilon^7 + 18(333\sqrt{9\epsilon^2 + 4} - 695)\epsilon^2 + 1660(\sqrt{9\epsilon^2 + 4} - 2)\epsilon + 232(\sqrt{9\epsilon^2 + 4} - 2) + 2916(7\sqrt{9\epsilon^2 + 4} - 25)\epsilon^6 + 243(100\sqrt{9\epsilon^2 + 4} - 341)\epsilon^5)}{\sqrt{9\epsilon^2 + 4}}.$$

Since $g'(\epsilon) = 0$ has five solutions, $\approx 0.023 - 0.665i$, $\approx 0.023 + 0.665i$, 0 , ≈ 0.295 , $\frac{1}{2}$, $g'(0.1) \approx 0.035 > 0$, $g'(0.4) \approx -0.078 < 0$, it follows that $g(\epsilon)$ is strictly increasing from $g(0) = 0$ to $g(\approx 0.295)$, strictly decreasing from $g(\approx 0.295)$ to $g(\frac{1}{2}) = 0$. Hence, if $0 < \epsilon < \frac{1}{2}$, $g(\epsilon) > 0$, $B'_{TB}(\epsilon) > 0$, which leads to the assertion of this theorem. $\square$

SI Dataset S1 (dataset_one.xlsx)

Raw data of average asymptotic bias of all estimators shown in Figure 1 in the main Text.

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